

Drone Meditations v1.1

Pre-App-Store-submission testing checklist · Build 7 (1.1) · iOS native app

Work through each section below before archiving and uploading to App Store Connect. Sections are ordered by risk (audio engine first since it had the most recent churn). Mark each checkbox as you complete it. Notes in italics give context — skip them if you already know the area.

Tip: use a fresh install (delete app + reinstall) or a second test device for the "first-launch" tests. iOS remembers permission state across installs from the same Apple ID.

1 · Audio engine — highest risk (heavy session churn)

■ **Fresh-install first launch:** NO mic permission prompt at launch · transport immediately responsive · welcome screens appear first

Recent fix moved mic prompt to first Listen tap; previously it fired at launch with a 5 s unresponsive transport.

■ **Play → Pause → Play:** smooth fade out, silent (no click), smooth fade in · no 5-second cutoff after resume
Click-free strategy leaves the engine running silently — no AU rebind, no click on click-free pause/stop hardware.

■ **Play → Stop → Play:** 10 s deep bloom fade · no click · reverb wash sustains in the middle · clean restart on Play

■ **Play → Stop → Stop → Stop:** rapid taps don't break anything

■ **Pause-after-Tune-to-Room (KNOWN ISSUE):** pause may feel longer than its 1.4 s spec (~3 s of audio bleed). Confirmed as iOS audio-buffer behavior, not a regression. Acceptable.

■ **Tune to Room:** opens · picks up hum / voice within 1-2 s · shows note name + Hz + cents · "Set as Root" works · Done → transport responsive

■ **Tune to Room while playing:** works during playback · doesn't crash · audio resumes cleanly on close

■ **Tune to Room → Done → Play → Pause → Play:** full transport cycle still works after Listen

■ **Record button:** full session records · mastering completes · share sheet appears with playable M4A · fade-in/out audible in the recording

■ **Lock screen / Now Playing:** pause/play from lock screen works

■ **Interruptions:** phone call · Siri · other audio app — audio resumes gracefully on return

■ **Background → foreground:** app survives suspension cycles

2 · v1.1 features (new this release)

■ **Quantize to scale:** toggle on a voice with S&H pitch LFO at depth=1, play. Verify arpeggio across chord notes (test C Minor 7, A Maj 9, Quantum Bells preset)

■ **Quantize persists across drift scene changes:** load Quantum Bells, change DRIFT pill to Ocean — quantize stays on per voice

■ **Multi-target LFOs:** enable PAN + CUT on one LFO, both targets respond to the same waveform

■ **New LFO shapes:** sawtooth + ramp work as expected

■ **Q (resonance) as LFO target:** produces audible resonance pumping

■ **FM index as LFO target:** produces FM modulation swell

■ **2 new presets:** Chord Arpeggio and Quantum Bells load and sound right

3 · iPad-specific (heavy layout changes)

- **iPad portrait:** strips at 1360 pt wide · breathe edge-to-edge · no overflows
- **iPad landscape:** same — full width usage
- **iPad transport:** single-row layout · all buttons fit + tappable
- **OSC strip on iPad:** waveform picker buttons big · LFO shape buttons match wave-picker size · target pills readable
- **No slider overlaps:** FILT row (BP → CUTOFF) · waveform row (PAN) · LFO rows
- **iPhone portrait still untouched** (regression check)
- **iPhone landscape:** full width · target pills don't overflow · transport compact
- **iPhone portrait LFO row:** 2-stacked sub-row layout · RATE / DEPTH labels horizontal (not vertical letter columns)

4 · iPhone-specific

- **Pinch-to-zoom Chladni** on main view works
- **Perform / fullscreen Chladni:** Exit button works · doesn't trap the user
- **Spectrum analyzer toggle:** no crash
- **Pill row scrollable** in portrait (CHORD / PRESET / DRIFT / etc.)
- **OSC nav pills** (1 / 2 / 3 / 4 row): scroll to selected strip

5 · State & persistence

- **Save per-voice preset:** name prompt sheet (no alert crackle) · saves · loads · deletes
- **Save global preset:** same flow, no audio click
- **User samples:** drop a WAV / MP3 / M4A via Files app → appears in OSC sample picker
- **Bundled samples:** all play correctly
- **Preset round-trip:** save a complex preset (LFOs + multi-target + quantize + drift), reload it, all fields restored exactly
- **Settings survive backgrounding:** drift state · master volume · current chord

6 · Manual + UI copy

- **dronemeditations.com/manual.html** loads · screenshots embedded · lightbox works
- **Manual link from app** (bottom-right Manual pill) opens in browser correctly
- **§3 Filter:** Q = resonance section reads correctly
- **§9 LFOs:** target list includes Q, FM, multi-target callout
- **§11 DRIFT:** quantize-to-scale subsection present
- No "What's new in v1.0" (renamed to Highlights)
- No old anatomy SVG diagram

7 · App Store submission build

- **CFBundleVersion = 7** (build 7) — already bumped
- **MARKETING_VERSION = 1.1** — already bumped
- **Privacy manifest** (PrivacyInfo.xcprivacy) accurate · NSPrivacyTracking=false · mic usage documented
- **Info.plist NSMicrophoneUsageDescription** reads sensibly (documents Tune to Room intent · no audio recorded or transmitted)

- **Debug prints wrapped in #if DEBUG** — already done (MicPitchDetector, AudioEngine, DroneController)
- **Release build** succeeds with no warnings: `xcodebuild -configuration Release`
- **Archive uploads** to App Store Connect cleanly

8 · Visual polish spot-check

- **Chladni renderer**: no boundary visible at any zoom level (full-screen pattern)
- **Master volume slider**: snappy · fades smoothly
- **Mini controls 2×2 grid** appears when controls hidden
- **Spectrum analyzer alternative view** renders without crash
- **Snapshot / photo export** works from Perform mode

9 · Final sanity

- **Install from TestFlight** (not just Cmd+R) — archive + upload + install via TestFlight on real iPhone. Catches things Debug builds miss
- **Fresh device** or wipe app data (Settings → Drone Meditations → delete data) — verifies first-launch flow
- **Old iPhone** (12 mini, SE, etc.) — performance check
- **All journeys (15+)** load and play through at least the first few stages
- **All chord types** in CHORD picker generate sensible frequencies
- **All tuning systems** (12-TET, Carlos $\alpha/\beta/\gamma$, Partch, Harrison) produce different audible tunings

10 · App Store Connect metadata (do these in browser)

- **What's New in This Version** — release notes for v1.1. Suggested copy:
 - Multi-target LFOs: drive pan + cutoff + pitch with a single modulator
 - New LFO shapes (sawtooth, ramp) and targets (resonance, FM index)
 - Quantize-to-scale per voice — drift snaps to the current chord
 - Trapezoidal stop fade with sustained reverb wash
 - Two new presets: Chord Arpeggio, Quantum Bells
 - iPad: full-width strips, tablet-scaled controls
 - Listen / Tune to Room reliability fixes
 - Click-free pause and stop transitions
- **Promotional text** (optional 170-char teaser) — keep v1.0 or update
- **Screenshots** — only re-upload if you want to highlight v1.1 features. v1.0 screenshots still accurate for core experience
- **Demo account** (App Review): not needed, no login flow
- **App Review notes**: mention mic is only used for "Tune to Room" (pitch detection, never recorded or transmitted) to preempt any reviewer questions

Known issues — acceptable for ship

- **Pause feels ~3 s after using Tune to Room** — iOS audio buffer behavior when the input route has been active. Not a regression, multiple fix attempts couldn't beat iOS
- **Play button has ~1-2 s audio latency** after first Listen use — same iOS audio-buffer behavior. Visual button response is immediate; just the actual audio onset lags